

|md2any

Markdown to PowerPoint, the fast way

Why this exists

Why this exists

Slide tools are heavy, hand-coded HTML decks rot, and PowerPoint refuses to die.

A single Rust binary that turns plain Markdown into a clean `.pptx` file is the right shape for the job. Edit in your favorite editor. Render in a heartbeat. Ship.

What you get

- Real `.pptx` output — opens in PowerPoint, Keynote, LibreOffice, Google Slides
- Light and dark themes, 16:9, 4:3, and portrait
- Auto-pagination of long content into continuation slides
- Side-by-side columns with `:::`
- Auto two-column layout for long lists
- Code listings with filenames, line numbers, language tags
- Bullets, nested lists, code blocks, tables, blockquotes
- Bold, *italic*, `inline code`, ~~strikethrough~~, [links](#)
- Front-matter YAML for deck metadata
- A single binary, no runtime dependencies



Layout features

Side-by-side columns

Put `:::` on a line by itself to split a slide horizontally.

Markdown

```
1  ## Slide title
2
3  Left column content here.
4
5  :::
6
7  Right column content here.
```

Rendered

Everything before the marker is the left column. Everything after is the right.

Side-by-side in practice

Why this matters

Two things you want to compare belong next to each other, not on consecutive slides. Side-by-side preserves the comparison; consecutive slides force the audience to remember the first half.

Use it for: before/after, cause/effect, do/don't, problem/solution, code/explanation.

How to use it

```
Left side  
:::  
Right side
```

Both columns paginate within themselves. You don't have to balance the heights manually — md2any scales each column to fit the available area independently.

Long lists auto-flow

Lists with more than 12 items wrap into two columns automatically. No marker needed.

- First item in the long list
- Second item with some text
- Third item demonstrating wrap
- Fourth item keeps going
- Fifth item adds variety
- Sixth item with **bold** text
- Seventh has `inline code`
- Eighth item is short
- Ninth item continues
- Tenth item is here
- Eleventh item — getting long
- Twelfth — last in the first column
- Thirteenth starts the second column
- Fourteenth item
- Fifteenth item
- Sixteenth item



Code listings

Inline language tag

A code fence with just a language gets a small language label.

```
rust
```

```
fn main() {  
    println!("Hello, slides!");  
}
```

With a filename caption

Put a filename after the language: ``rust src/main.rs. The filename appears in a header bar above the code, with the language tag to its right.

src/main.rs · rust

```
1  use anyhow::Result;
2  use clap::Parser;
3
4  #[derive(Parser)]
5  struct Cli {
6      input: String,
7  }
8
9  fn main() -> Result<()> {
10     let cli = Cli::parse();
11     println!("converting {}", cli.input);
12     Ok(())
13 }
```

With line numbers

Code blocks longer than five lines get line numbers automatically.

pipeline.py · python

```
1  def transform(rows):
2      for row in rows:
3          row["price"] = row["price"] * 1.1
4          row["currency"] = "EUR"
5          yield row
6
7  def write_jsonl(rows, path):
8      with open(path, "w") as f:
9          for row in rows:
10             f.write(json.dumps(row) + "\n")
```



Layouts

Four ways to dress the same deck

md2any ships with four built-in layouts. Pick one with `--layout` or set `layout:` in front matter. Each one is a coordinated treatment — background, title style, section dividers, and title slide all change together.

- `clean` — minimal title underline, full-bleed content (default)
- `studio` — vertical accent rail, italic titles, big background numerals on sections
- `frame` — left sidebar holds deck title and slide number; content lives on the right
- `bold` — full-width accent block behind every title

Same markdown, four very different decks. Light and dark themes apply to all four.

bash

```
md2any talk.md --layout studio
md2any talk.md --layout frame --theme dark
md2any talk.md --layout bold
```

When to use which

clean — talks, internal updates, anything where you want the content to be the loudest thing on the slide.

studio — editorial-feel decks, design reviews, fund-raising memos. The rail and numerals look serious without being heavy.

frame — long-form decks where the audience benefits from a persistent reference panel (chapter name, page number). Good for training and onboarding decks.

bold — keynote-style talks where every section transition should feel like an event. Lots of accent color on every slide.



Mainframe languages

COBOL

Case-insensitive, hyphens in identifiers, column-7 comment indicator.

HELLO.cbl · cobol

```
1      IDENTIFICATION DIVISION.  
2      PROGRAM-ID. HELLO.  
3      *  
4      DATA DIVISION.  
5      WORKING-STORAGE SECTION.  
6      01 WS-NAME          PIC X(20) VALUE 'mainframe'.  
7      01 WS-COUNT        PIC 9(3)  COMP-3 VALUE 5.  
8      *  
9      PROCEDURE DIVISION.  
10     DISPLAY 'Hello, ' WS-NAME '!'.  
11     PERFORM VARYING I FROM 1 BY 1 UNTIL I > WS-COUNT  
12         DISPLAY 'iteration ' I  
13     END-PERFORM.  
14     STOP RUN.
```

`/**` for comments, `//` starts every statement, parameters are `KEY=VALUE`.

HELLO.jcl · jcl

```
1  //HELLO      JOB  (ACCT), 'PAUL', CLASS=A, MSGCLASS=X, NOTIFY=&SYSUID
2  /**
3  /** RUN A SIMPLE BATCH STEP
4  /**
5  //STEP1      EXEC PGM=IEFBR14
6  //SYSPRINT DD  SYSOUT=*
7  //SYSIN      DD  *
8  HELLO MAINFRAME
9  /*
10 //STEP2      EXEC PGM=SORT, COND=(0, LT, STEP1)
11 //SORTIN     DD  DSN=PAUL.INPUT, DISP=SHR
12 //SORTOUT    DD  DSN=PAUL.OUTPUT, DISP=(NEW, CATLG)
```

REXX

`/* */` block comments, case-insensitive keywords, classic scripting feel.

hello.rex · rexx

```
1  /* hello.rex – sample REXX script */
2  parse arg name
3  if name = '' then name = 'mainframe'
4  say 'Hello,' name'!'
5  do i = 1 to 5
6      say 'iteration' i
7  end
8  exit 0
```

PL/I

`/* */` comments, single-quoted strings, declarations with attributes.

hello.pli · pli

```
1  HELLO: PROC OPTIONS(MAIN);
2      DCL NAME          CHAR(20) VARYING INIT('mainframe');
3      DCL COUNT          FIXED BIN(31) INIT(5);
4      DCL I              FIXED BIN(31);
5
6      PUT SKIP LIST('Hello, ' || NAME || '!');
7      DO I = 1 TO COUNT;
8          PUT SKIP LIST('iteration ', I);
9      END;
10  END HELLO;
```

HLASM

Column-1 * for comments, mnemonics highlighted, &VAR macro variables.

HELLO.asm · hlasm

```
1  *  HELLO WORLD ASSEMBLER PROGRAM
2  HELLO  CSECT
3          USING  HELLO,R15
4          STM   R14,R12,12(R13)    SAVE  REGISTERS
5          LA    R3,5               ITERATION COUNT
6  LOOP   EQU   *
7          WTO   'Hello, mainframe!' WRITE TO OPERATOR
8          BCT   R3,LOOP            DECREMENT AND BRANCH
9          LM    R14,R12,12(R13)    RESTORE REGISTERS
10         BR    R14                RETURN
11         LTORG
12         END   HELLO
```

DB2 DDL

Recognises the standard SQL keywords plus DB2 catalog and tablespace specifics.

DB2 DDL (cont.)

CUSTOMER.ddl · db2

```
1  -- CREATE A TABLESPACE AND A TABLE INSIDE IT
2  CREATE TABLESPACE TS001
3      IN DATABASE DBPROD
4      USING STOGROUP SG1
5      PRIQTY 720 SECQTY 720
6      BUFFERPOOL BP0
7      LOCKSIZE PAGE
8      COMPRESS YES;
9
10 CREATE TABLE PAUL.CUSTOMER (
11     CUST_ID      INTEGER          NOT NULL GENERATED ALWAYS AS IDENTITY,
12     CUST_NAME    VARCHAR(50)      NOT NULL,
13     CUST_TIER    CHAR(1)          DEFAULT 'B' CHECK (CUST_TIER IN ('A','B','C')),
14     CREATED_AT   TIMESTAMP        NOT NULL DEFAULT CURRENT_TIMESTAMP,
15     PRIMARY KEY (CUST_ID)
16 ) IN DBPROD.TS001
17     CCSID UNICODE;
18
```

DB2 DDL (cont.)

db2

```
1 CREATE INDEX PAUL.CUST_NAME_IX
2   ON PAUL.CUSTOMER (CUST_NAME ASC)
3   BUFFERPOOL BP1;
```




Themes

Light vs dark

Light theme

Default. White background, slate text, sky-blue accent. Reads well in a bright room, prints clean, looks like a polished tech-company deck.

Use `--theme light` or omit (it's the default).

Dark theme

Slate-900 background, frost-bright accent. Reads well on stage, easier on the eyes during a long talk, photographs cleanly from the back of the room.

Use `--theme dark`.



Aspect ratios

Choose with --aspect

```
bash
```

```
md2any talk.md --aspect 16:9      # default, widescreen
md2any talk.md --aspect 4:3       # legacy projectors
md2any talk.md --aspect 9:16      # portrait, phone-shaped
md2any talk.md --aspect a4        # paper-shaped portrait
md2any talk.md --aspect letter    # US letter portrait
```

The portrait layouts adjust typography automatically — smaller titles, narrower content area, single-column lists.



Images

Embedded images

Use standard markdown image syntax. PNG and JPEG are supported. Paths resolve relative to the markdown file.

Embedded images (cont.)



Hero image showing the md2any wordmark on a gradient background

Side-by-side image + text

Images compose with everything else, including columns. Put an image on one side, your commentary on the other.



A round badge labeled PNG + JPEG

Two supported formats

PNG and JPEG. No GIF, no SVG, no WebP for now.

Sizing

Images fit to the available content area, preserving aspect ratio. Auto-centered horizontally.

Captions

Whatever you write in the markdown alt text becomes a small caption below the image.



Speaker notes

How to add notes

Put an HTML comment that starts with `notes:` or `speaker notes:` anywhere on the slide. They attach to the slide they're written on.

markdown

```
## A slide title
```

```
Visible content here.
```

```
<!-- notes: Reminder to ad-lib about the migration timeline. Reference the Q3 deck. -->
```

The notes appear in PowerPoint's presenter view and printed as note pages.

Multiple notes

Multiple `<!-- notes: -->` comments on the same slide are concatenated with blank lines between them. Useful when notes accrete during drafting.

Tables

Feature matrix

Feature	Status	Notes
Headings	Done	H1 title, H2 content, H3+ inline
Lists	Done	Nested, auto-2-column when long
Code blocks	Done	Filename, language, line numbers
Tables	Done	Banded rows, accent header
Side-by-side	Done	::: marker
Portrait	Done	9:16, A4, Letter
Dark mode	Done	Polished palette
Images	TODO	v0.2
Speaker notes	TODO	v0.2



Quotes

Quotes

The best presentation tool is the one that gets out of your way.

Plain text in, polished slides out. That's the whole product.

Quotes

This is a continuation slide created with an explicit - - - rule. The title carries over.



Performance

Performance

Single-pass Markdown parse via `pulldown-cmark`. OOXML emitted directly to a `zip::ZipWriter`. No intermediate AST beyond the slide IR. Cold-start binary in tens of milliseconds.

A thousand-slide deck renders in about 30 ms on commodity hardware.

That's it

That's it

Write Markdown. Run one command. Open the `.pptx`. Present.